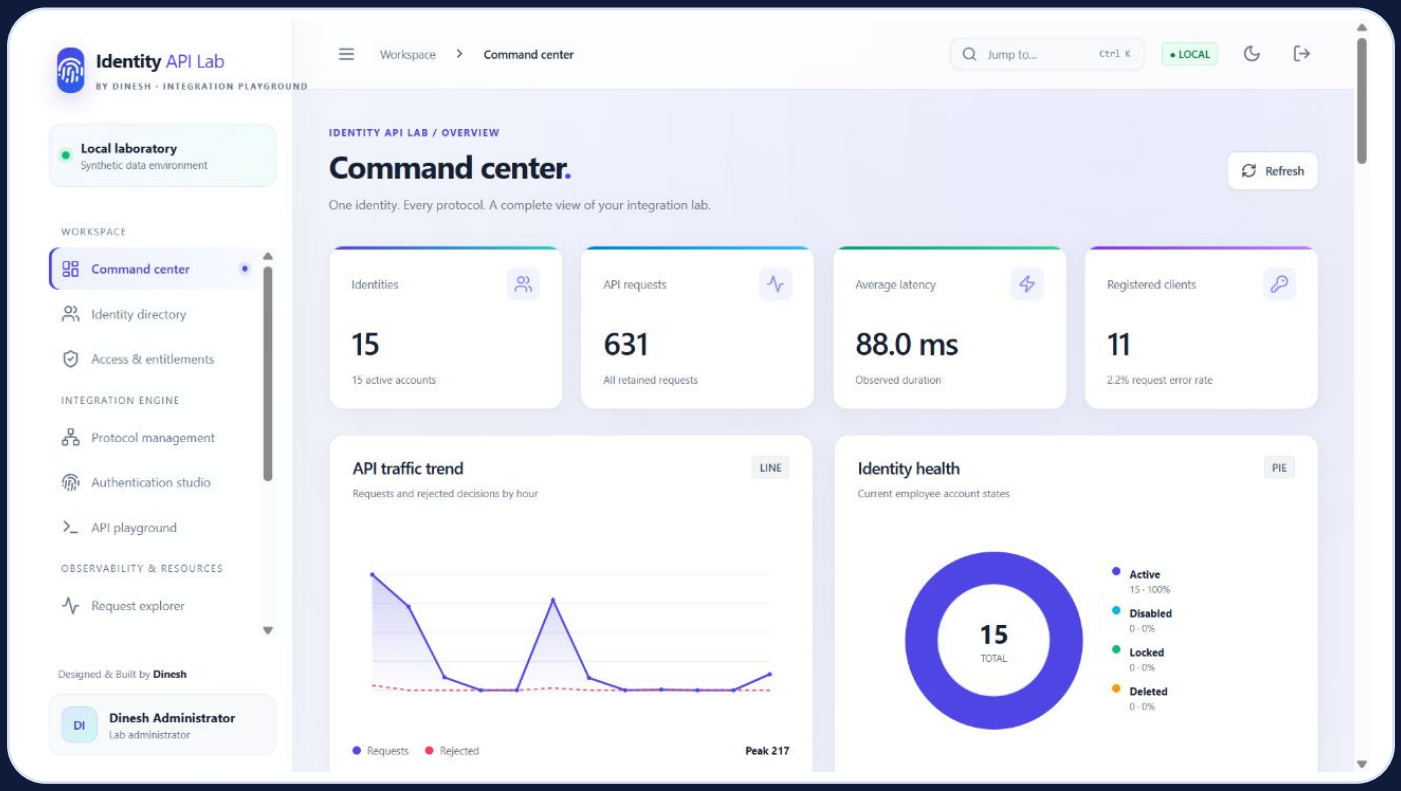
PRODUCT BROCHURE

Identity API Lab

One environment for identity integration testing

Protocols, authentication, permissions, banking workflows and evidence share one canonical identity model.



Designed & Built by Dinesh

github.com/dmenh/identity-api-lab

PRODUCT OVERVIEW

A full integration test, not disconnected sample APIs

Every interface works against the same identities, access policy, sessions and audit records.

01

Register

Choose authentication methods, protocols, scopes and attributes.

02

Provision

Create an employee through SCIM, REST, OData, gRPC or CSV.

03

Authorize

Assign roles and calculate effective permissions from live state.

04

Operate

Show the employee only the banking modules they may use.

05

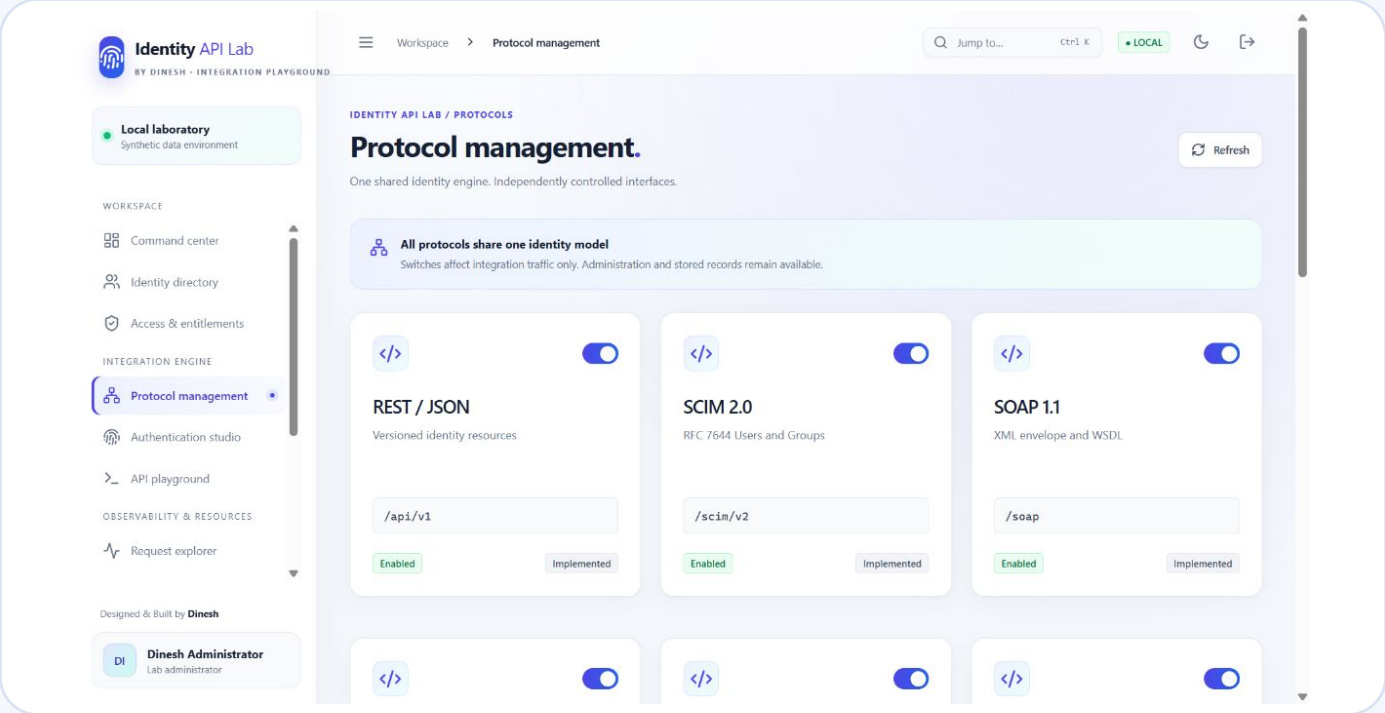
Inspect

Review request timing, authentication decisions and audit evidence.

PROTOCOL MANAGEMENT

Eleven adapters. One identity state.

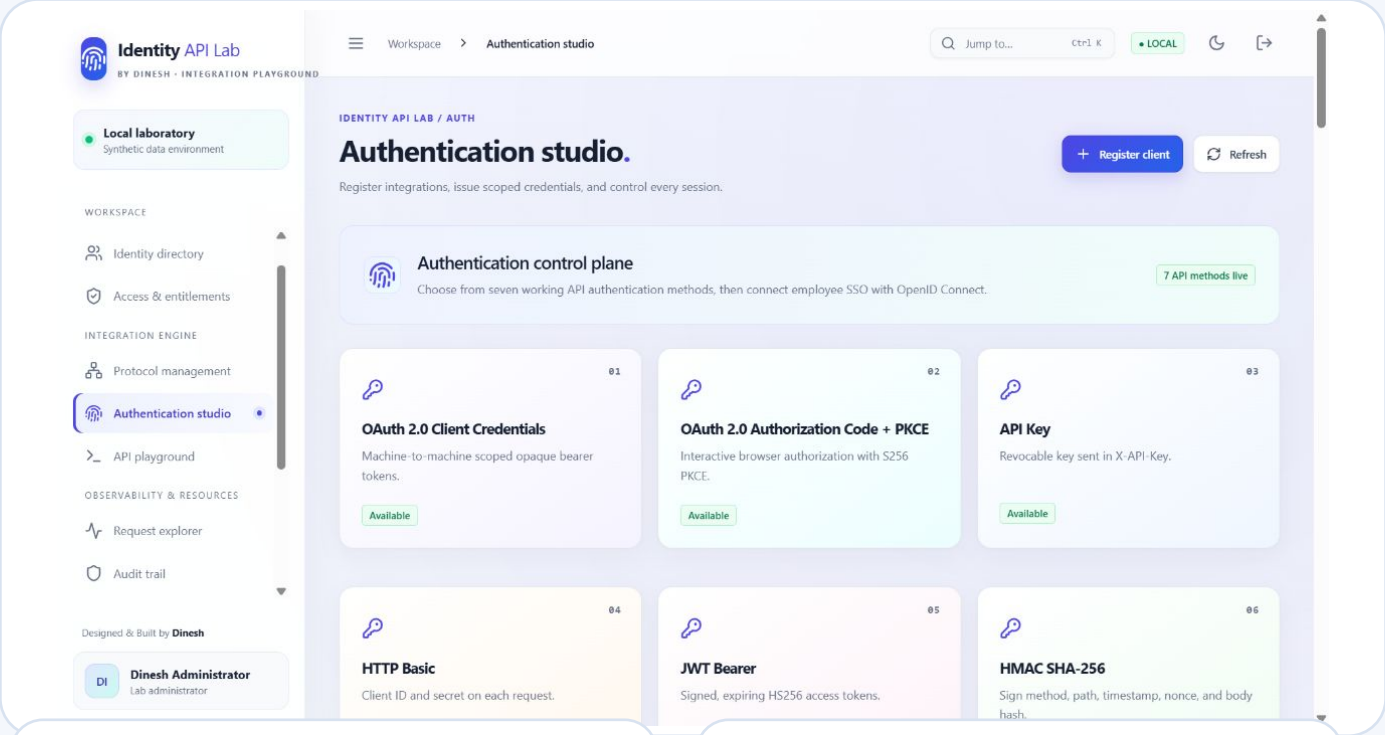
Enable each interface independently while administration and canonical records remain available.



- REST
- SCIM 2.0
- SCIM 1.1
- SOAP 1.1
- SOAP 1.2
- GraphQL
- OData 4.0
- gRPC
- JSON-RPC
- Webhooks
- CSV / SFTP

AUTHENTICATION STUDIO

Credentials, scopes and attribute policy in one place



OAuth 2.0

Client Credentials and Authorization Code with S256 PKCE

API credentials

API key, HTTP Basic and HTTP Digest

Signed access

JWT bearer and HMAC SHA-256 request signing

Immediate control

Rotate credentials, revoke tokens and terminate sessions

ACCESS & ENTITLEMENTS

Permissions change both API behavior and the employee interface

Custom permissions, roles and groups produce effective access that the server enforces on every request.

Role and group model

Permissions define individual operations. Roles collect permissions for a job. Groups grant shared roles to their members.

Attribute-level policy

Every API client has separate readable and writable identity fields. Requests outside that policy receive a controlled error.

Example policy

READABLE

username **email** **displayName**

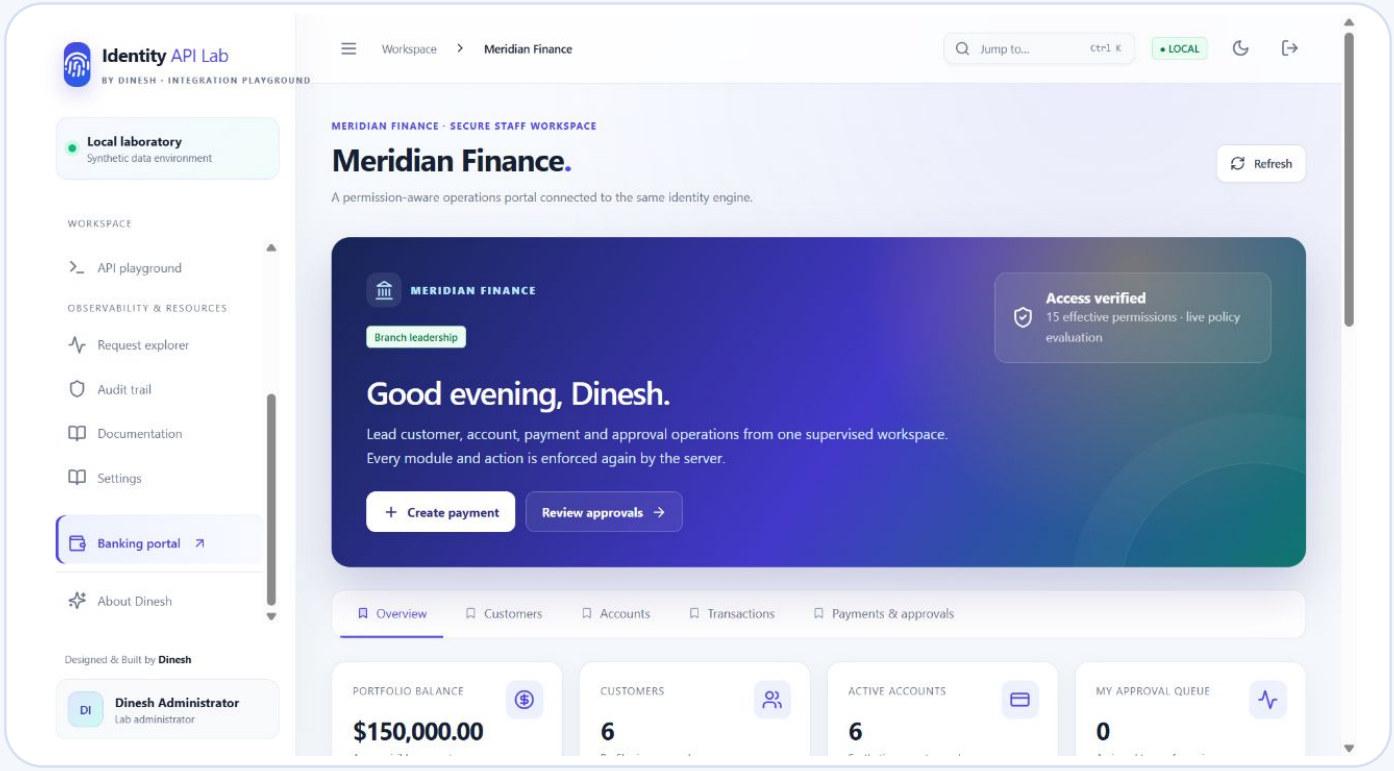
WRITABLE

displayName **department** **jobTitle**

Admin testing

View the portal as another employee, compare the interface, then return through the persistent preview banner. The switch is audited and never reveals the employee password.

Effective permissions become visible business behavior



1. Maker creates payment
2. Eligible checker reviews
3. Balances and audit post

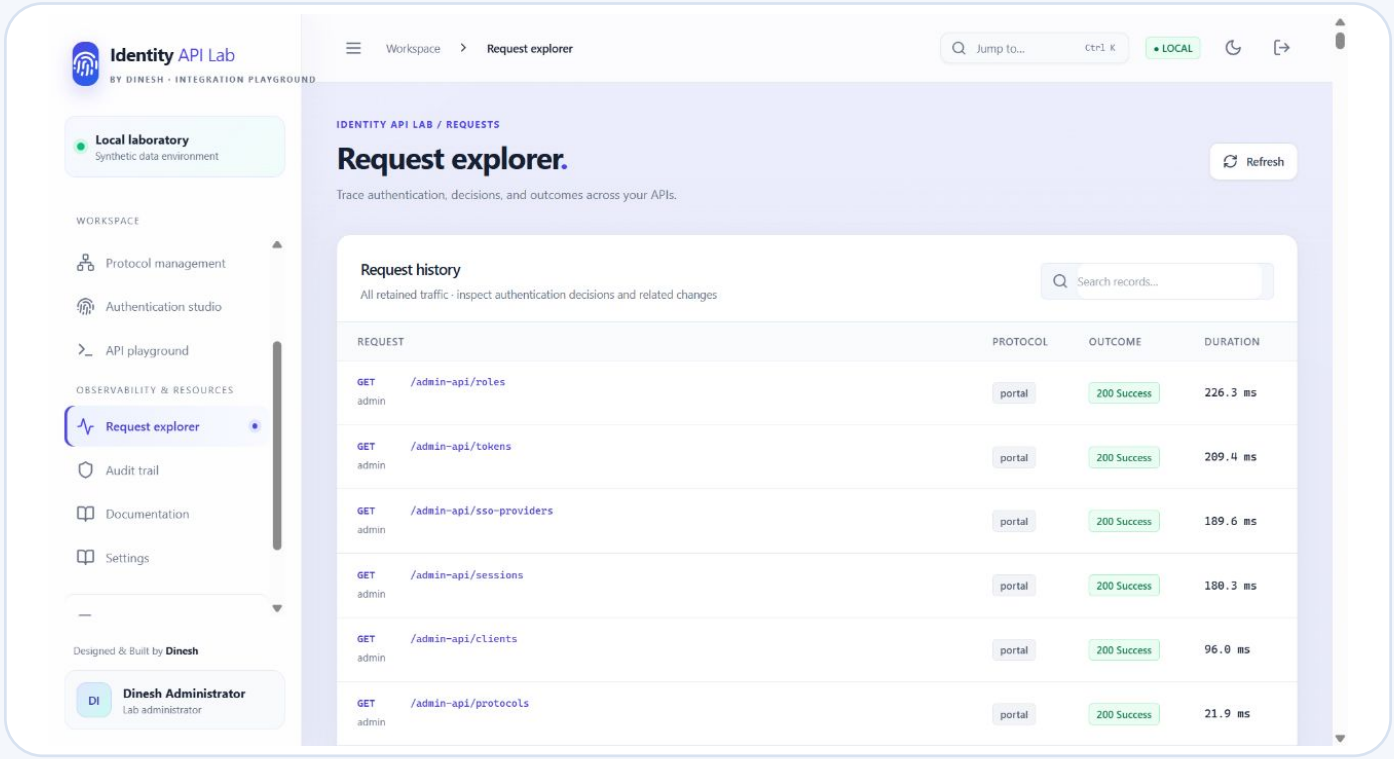
Maker/checker enforcement

The initiator cannot approve the same payment. Approval checks the assigned reviewer, permissions, account state and available balance before posting.

REQUEST EXPLORER

Every decision leaves evidence

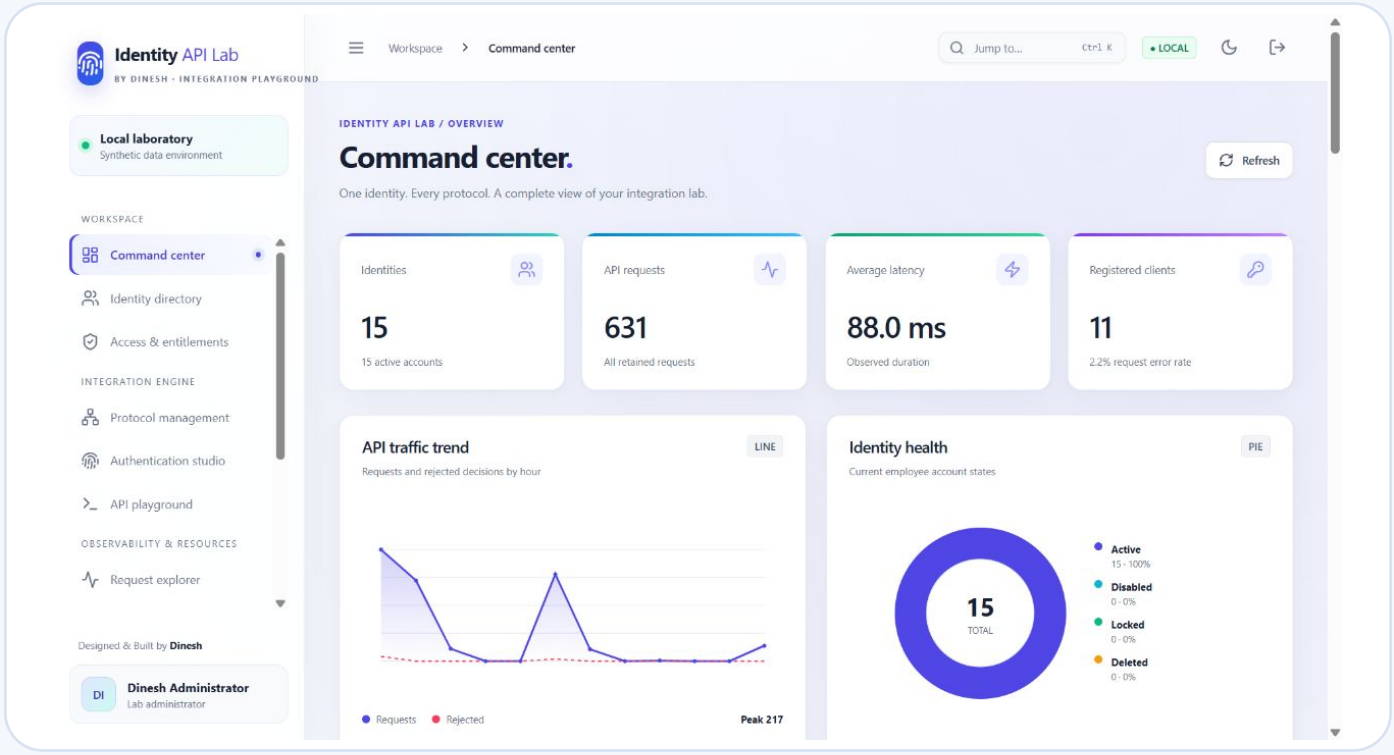
Trace the transport request to the authentication result, identity change and business audit event.



1. Request and correlation IDs
2. Actor and authentication method
3. Allowed or rejected decision
4. Observed execution duration
5. Related identity and audit changes

COMMAND CENTER

Live dashboards explain traffic, sessions and risk



15 synthetic employees

15 countries

11 protocol adapters

7 client auth methods

Line · Bump · Calendar · Choropleth · Pie · Heatmap · Sunburst · Bar

DEPLOYMENT

Start locally, then publish isolated test environments

The repository includes Docker Compose, Railway configuration, a Render blueprint and repeatable health checks.

Local Docker

One command builds the application, creates secrets, initializes data and starts PostgreSQL. MySQL and SQLite are also available.

Private server

Add TLS termination, managed secrets, backups, outbound restrictions and centralized audit storage.

Hosted tenant

Provision one isolated application and database pair per customer. Add billing, quotas, retention and lifecycle automation.

Recommended hosted product path

1. **Connect the GitHub repository to Railway.**
2. **Add managed PostgreSQL and required secrets.**
3. **Publish a reusable template after the first verified deployment.**
4. **Suspend idle trials and charge for persistent isolated environments.**

Native gRPC

Public gRPC requires a platform TCP endpoint. HTTP endpoints use the normal HTTPS application domain.

Build confidence before connecting a real identity platform

Use a realistic employee lifecycle, permission-aware banking portal and complete request evidence in one lightweight lab.

Repository

github.com/dmenh/identity-api-lab

Contact

linkedin.com/in/dineshwarmanukonda

Support the project

Sponsorship funds protocol compatibility, security testing and documentation.

Designed & Built by Dinesh

Synthetic data only. No real banking services or customer records.